→ look at scribe data

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⇒ ESA (produces a judgement of relevance btw. any 2 words)

- semantic interpretor - It computes the semantic relatedness btw. words & docs. by representing them in a high-dim. concept space derived from Wiki. We represent text as a vector of explicit human-defined concepts. It is easier to interpret because each dim. is a real Wiki article.
- sample text fragments (conceptualizaⁿ)
  In ESA, conceptualizaⁿ means the process of converting a short text fragment into a weighted vector of explicit concepts.
  Concept space instead of word space.

- evaluaⁿ (spearman rank correlaⁿ) → human relevance score
  Pearson coefficient will be only able to capture the linear dependencies.
  → preserve the human judgment & ESA results.

$n \Rightarrow$ no. of word pairs

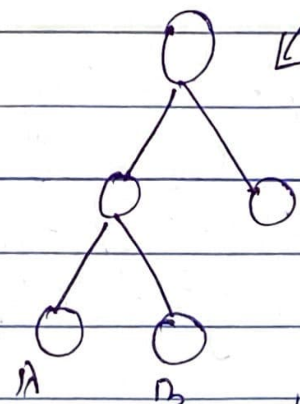
$d_i \Rightarrow$ diff. btw. ranks of each pair

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$$

$$\rho \in [-1, +1]$$

→ <u>WordNet</u> (informa" theoretic measures), path-based measures)

↳ find the path len., closer the synsets are




we can look at specificity of the nodes.

sim (A, B) = sim (C, D)

⇒ based on the path-based measures.

but this is not always correct. How to check the specificity btw. any 2 nodes?

Info. Content ← $\boxed{IC(\text{node}) = \log \frac{1}{P(c)}}$

that means probab. of root will be 1 & IC of root would be 0.

this is based on the trick of counting.

Resnik similarity - Semantic similarity measure based on IC in taxonomy like WordNet

IC (LCS)

JC distance - extends resnik similarity.

It measures how far 2 concepts are apart in terms of IC.

$IC(c) = -\log P(c)$

Lin Measure → provides a normalized score

(formulla next page)

Lowest common ancestor

$$\frac{2\, IC(LCA(c_1, c_2))}{IC(c_1) + IC(c_2)}$$

- Distributional Word Similarity - We try to infer the meaning of few words (difficult) rather than looking at dic., we can just infer abt. the same by context.
Two words are similar if they have a similar context.
Word Across docs, term content matrix.
Two words are similar if their content vectors are similar.

word X word matrix

→ has info.

- PPMI (Positive Pointwise Mutual Info.)
(instead of TF-IDF) → for bigrams

$$PMI(x, y) = \log_2 \frac{P(x, y)}{P(x)\, P(y)}$$

PPMI = PMI values for less than 0 get replaced by 0.
↓
Weighing scheme in distributional semantics.
It measures how strongly a word is associated with a context word as compared to chance.